King Saud University
College of Computer and Information Sciences
Department of Information Technology
IT 362 – Data Science
2st Semester 1447 H

Tech Job Skills Analysis in Saudi Arabia

Logbook

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Phase	Task	Date	Decisions & Rationale	Challenges & Solutions	Implementation
1: Data Collection Research and Assessment	API	4 Feb	We collected job data using the JSearch API instead of web scraping to ensure structured, reliable, and consistent data.	We faced API quota limitations and addressed them by optimizing query parameters and adjusting the number of retrieved pages.	Python (requests + pandas) was used to retrieve and normalize the data. The final dataset includes 144 job records with 31 features.
	Dataset Expansion	12 Feb	We increased the number of queries and pages to improve dataset size and diversity.	API quota limitations required optimizing query parameters.	Additional job roles were added to the search list, resulting in 1,024 records.
	Data Quality Assessment	13 Feb	We reviewed the dataset to ensure it was correctly loaded and properly formatted before proceeding with analysis.	During the review, we identified encoding issues affecting the readability of some text fields. The issue was resolved by adjusting the file encoding to UTF-8.	The dataset was reloaded and verified after updating the encoding format to ensure correct text representation.
	Literature Review	8 Feb	We searched for academic studies related to skill extraction from job advertisements.	Some studies were not relevant, so we refined the search keywords.	Selected relevant papers and summarized their problem, dataset, methods, and findings.

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2: Data Processing, Cleaning, and Exploratory Data Analysis (EDA)	Data Processing & Cleaning	23 Feb	The dataset was reviewed for missing values, duplicates, and irrelevant features. Columns with 100% null values were removed, and job descriptions were normalized through text preprocessing.	Several empty columns and four duplicate records were identified. These were removed to improve data quality and consistency.	Python (pandas and re) was used to inspect null values, drop unnecessary columns, remove duplicates, and preprocess text. The cleaned dataset (1,020 records, 22 features) was exported for further analysis.
	Primary Data EDA	26 Feb	We performed exploratory data analysis on the primary dataset (that are collected from Saudi job postings). The goal was to better understand the dataset and identify patterns in job roles and required skills.	While analyzing the data, we noticed that some job titles had different formats or capitalization. To make the analysis more accurate, the titles were cleaned and grouped into main roles such as Data Analyst, Data Engineer, and AI/ML Engineer.	Python was used for the analysis, mainly using pandas and matplotlib. The analysis included examining job role distribution, identifying the most common skills, calculating skill percentages, and comparing skills across different roles.
	Secondary Data EDA	1 March	A secondary dataset from Kaggle containing data science job postings was selected to complement the primary dataset. The dataset was chosen because it includes relevant variables such as job title, and required skills allowing comparison with the primary data collected from the Saudi job market.	Some variables contained missing values and the skills column included list-formatted text that required preprocessing. These issues were addressed by inspecting missing values, cleaning the data, and extracting skills for analysis.	Python (pandas, matplotlib, seaborn) was used to perform exploratory data analysis. Visualizations were generated to analyze job title distribution, seniority levels, most common skills, and geographic distribution of job postings.

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Phase 3 Modelling and Communication	Data Preparation & Feature Engineering	15 Apr	Prepared the dataset for classification by selecting relevant features, cleaning and standardizing job titles, and grouping them into four main roles to simplify the prediction task.	Job titles contained many variations and irrelevant roles, leading to class imbalance. This was addressed by cleaning titles, grouping similar roles, and filtering only data and AI-related positions.	Used Python (pandas, re) to clean job titles, create role labels, filter relevant classes, and prepare the text feature for modelling.
	Model Building	21Apr	Built three classification models to predict job roles from the TF-IDF features. Logistic Regression was used as a baseline because it is simple and easy to interpret. Decision Tree was included to capture non-linear patterns, and Random Forest was used to improve performance by combining multiple trees.	The main challenge was class imbalance between job roles. This was handled using class_weight='balanced' in all models. Another issue was potential overfitting in Decision Tree, which was reduced by limiting the tree depth.	Implemented the models using scikit-learn in Python. Each model was trained on X_train and y_train, then evaluated on X_test. Performance was compared using accuracy and F1-score to select the best model.